Viren Khandal

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Experience

Tesla — Fremont, CA

Tech Lead, Generative AI – Inference & GPU Engineering — *Apr 2026 – Present*

- Spearheading Tesla's generative AI inference gateway, serving **30M+ requests** and **~2T tokens** daily across LLM and multimodal models, and setting direction for serving, GPU performance, and RL post-training.
- Architected on-prem GPU infrastructure saving **~$10M/year**, including a SLURM cluster co-scheduling training and inference, a Mooncake-based KV cache store, and rollout pipelines for LLM RL post-training.
- Directing NVIDIA and AMD partnerships that achieved **2.8× throughput** over baseline serving stacks on Tesla's hardware through custom inference recipes and kernel engineering.

Senior Machine Learning Engineer – Manufacturing AI — *Dec 2022 – Apr 2026*

- Led ML development across 4 global factories, spanning vision-guided robotics, large-scale defect detection, and core infrastructure for an internal computer vision training platform.
- Developed few-shot domain adaptation and knowledge distillation methods to adapt large pretrained vision models to new factory domains and compress them for real-time edge inference.
- Shipped production computer vision systems for automated quality inspection on high-throughput lines.

Ponder (Stealth) — Remote

Co-Founder — *Oct 2024 – Jan 2026*

- Led full backend and ML development for an AI app that generates multi-perspective audio podcasts from any user query, with each perspective grounded in cited research and credible sources.
- Architected retrieval-augmented generation pipelines to synthesize web-scale information into structured, multi-viewpoint audio content with source attribution.
- Built end-to-end backend infrastructure including API design, audio generation, and serving systems to support real-time podcast creation at scale.

University of California, Berkeley – VeHICaL — Berkeley, CA

Research Engineer — *May 2021 – Dec 2022*

- Led research on safe autonomous-to-human perception handoffs under Professors Sanjit Seshia and Björn Hartmann (vehical.org), formalizing stochastic models for verified human-autonomy interaction.
- Developed VR-based human-in-the-loop experimental tools to study driver takeover scenarios with high ecological validity.

Stanford Autonomous Systems Lab — Stanford, CA

Visiting Researcher — *Apr 2022 – Dec 2022*

- Investigated out-of-distribution (OOD) events in human-autonomy interaction under Professor Marco Pavone (stanfordasl.github.io), developing frameworks for safe real-time control handoffs during OOD scenarios.

Education

University of California, Berkeley — Berkeley, CA

B.A. Computer Science & Applied Mathematics (Double Major) — *Aug 2019 – May 2022*

- Specialization in Machine Learning and Artificial Intelligence
- Coursework: Deep Learning, Machine Learning, Artificial Intelligence, Efficient Algorithms, Data Structures, Statistical Prediction, Numerical/Real/Complex Analysis, Advanced Linear Algebra, Abstract Algebra

Stanford University — Stanford, CA

Certificate in Computer Science — *May 2018 – Sep 2018*

- Coursework: Data Mining and Analysis, Client-Side Internet Technologies

Technical Skills

Languages:	Python, C/C++, Go, Java, JavaScript, SQL
ML/AI:	PyTorch, TensorFlow, OpenCV, LLMs, RL Post-Training (RLHF, GRPO), RAG
Inference/GPU:	LLM Serving, vLLM, CUDA, ROCm, SLURM, Mooncake
Infrastructure:	AWS, GCP, Docker, Kubernetes